

CSC1016

Group 7 Project 2

**System Analysis & Process Design for the
University Parking Permit & Appeals Service**

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1. Introduction

This report presents the system analysis and design for the University Parking Permit & Appeals Service originally proposed by Group NLSC in Project 1. Our team has translated the problem description into a detailed, executable process landscape using BPMN 2.0 and supporting UML artefacts. Where the original proposal left requirements with room for improvement. We redesigned and improved the system and design.

1.1 Group Members & Contributions

Members	Role	Contribution	Estimated %
Kiran Joseph 40699	Project Lead & System Analyst	Scoped the analysis, led the client's liaison with Group NLSC, authored the process landscape and drafted Sections 1, 2 and 5 of the report.	- 2356800541%
Andrey Pricop 18312	BPMN Designer 1 (Process 1, Process 2, Process 6)	Designed the advanced BPMN collaboration diagrams for Permit Application (Process 1), Payment Processing (Process 2) and Permit Renewal (Process 6); introduced event-based gateways and boundary events for deadlines.	-234790754%
Patrick Sebastian 21936	BPMN Designer 2 (Process 3, Process 4, Process 5) & UI Mockups	Designed the BPMN for Permit Issuance (Process 3), Violation Recording (Process 4) and Fine Appeal (Process 5); produced the optional HTML/CSS form wireframes.	- 6348943578%
Jake George 35982	System Designer & Report Assembly	Modelling the UML class diagram and data structures, reviewed consistency between BPMN and UML, formatted the report, and performed quality review of all BPMN files.	∞

1.2 Description of the Client System

Group NLSC proposed a digital University Parking Permit & Appeals Service for a university that previously operated using a largely manual, paper-based system. The core problems identified by them included lost application emails, manual payment reconciliation errors, no automated enforcement of business rules, and a lack of a clear audit trail for permit appeals.

The proposed system spans six interconnected core processes covering the full parking lifecycle: permit application and eligibility checking, payment processing, permit issuance and zone allocation, parking fine and violation recording, fine appeal and resolution, and permit renewal and expiry management.

Key business rules defined by the client include restricting permits to registered students and staff with valid university IDs, blocking issuance where unpaid fines exist, enforcing zone allocation priorities (Staff are Zone A, Students go to Zones B-D), requiring payment within 48 hours of

approval, enforcing a 14-day appeal submission window and maintaining a maximum capacity of 500 active permits at any time. The client's success criteria include reducing application processing time from 5 days to under 1 working day, eliminating manual payment reconciliation errors and ensuring all appeals receive a written outcome within 7 days.

Our design expands this into a complete process landscape with explicit exception handling, event driven timers for all deadlines, collaboration across multiple organizational participants and a supporting data model.

2. Process Landscape

Our process landscape breaks down the client's 6 processes into a set of designed processes (Process 1- Process 6) that together cover the full lifecycle. Each designed process maps back to the corresponding original process proposed by the original group and each has been enriched with advanced BPMN elements including collaboration pools, timer and message events, subprocesses, event-based gateways, boundary events and typed tasks making it a realistic and operationally complete model.

Designed Process	Related Original Process	Description
Process 1: Permit Application & Eligibility Verification	Process 1: Permit Application & Eligibility Check	Handles application submission, completeness validation, automated eligibility checks (ID, unpaid fines, active permit rule) via automated service tasks and an approval/rejection notification.
Process 2: Invoice Generation & Payment Processing	Process 2: Payment Processing	Generates the invoice upon approval, processes online and in-person payments through integration with the university's finance system and enforces the 48-hour reminder and 5-day cancellation rules using boundary timer events on a wait task.
Process 3: Zone Allocation & Digital Permit Issue	Process 3: Permit Issue & Zone Allocation	Applies zone priority rules (Staff → Zone A, Students → Zones BD), reserves a specific bay, generates the Digital Permit ID in parallel with updating the master database and emails the permit.
Process 4: Violation Detection & Fine Issuance	Process 4: Parking Fine & Violation Recording	Supports ground enforcement by scanning a plate, querying the permit database, capturing photographic evidence and automatically logging and emailing the fine notice. Also handles escalation for unregistered vehicles.
Process 5: Fine Appeal & Resolution	Process 5: Fine Appeal & Resolution	Manages appeal submission with a 14-day timer, waiting for the weekly panel meeting

		(modelled as a 7-day timer) and records the decision (cancel, reduce or uphold) in finance records before notifying the appellant.
Process 6: Permit Renewal & Expiry Management	Process 6: Permit Renewal & Expiry Management	Triggers automated 30-day and 7-day reminders, re-checks eligibility via automated service tasks, handles payment and new permit issuance or deactivate the permit on expiry without renewal.

3. Process Design

3.1 Process Models

For each of the designed processes below we have provided:

- The BPMN diagram exported from our model
- A narrative of the advanced BPMN elements used
- Design justification including how the design improves the original proposal, assumptions taken, and open clarifications.

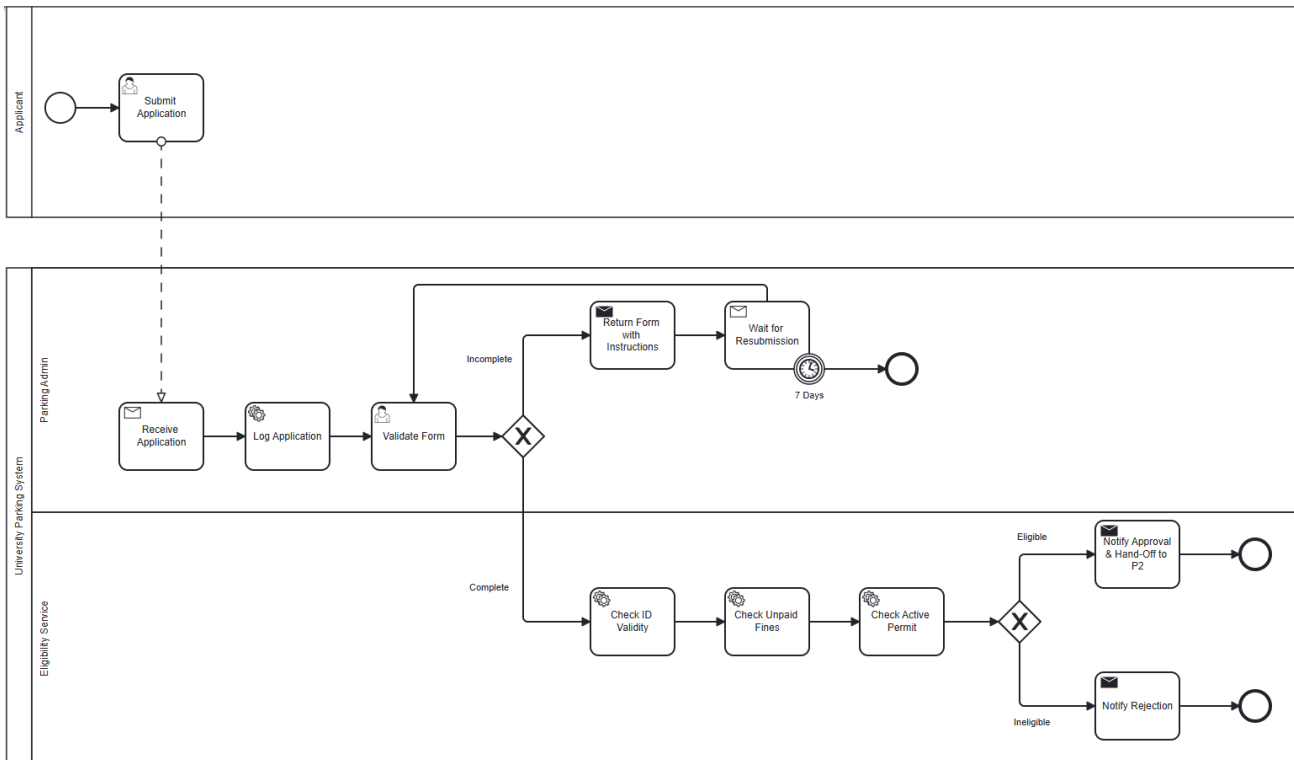


Figure 1: Process 1 - Permit Application & Eligibility Verification

BPMN Elements used:

Pools and lanes: Applicant pool and University Parking System pool with two lanes: Parking Admin, Eligibility Service.

Start Message Event: "Application received" in the University Parking System pool triggered by a message flow from the Applicant pool.

User Tasks: "Validate form completeness".

Service Tasks: "Check ID validity", "Check unpaid fines" and "Check active permit" for automated eligibility checks.

Exclusive Gateways:

- After "Validate form". Branches: "Complete" / "Incomplete".
- After eligibility checks. Branches: "Eligible" / "Ineligible".

Send Tasks: "Return form with instructions", "Notify rejection" and "Notify approval & hand-off to P2".

Receive Task: "Wait for resubmission" with a **timer boundary event (7 days)**. When timeout occurs it's a rejection. When message occurs, it loops back to validation.

End Events: Plain end events for rejection and approval paths.

Design justification:

The process is designed to make clear what the Permit Office is responsible for and what the citizens must do. Since the citizen submits a form and waits for a response, the process uses messages (such as email notifications) instead of real-time steps. The process has decision points after checking the form and after checking for an existing permit then it follows one of two possible paths depending on the validity of the information. Also, a 7 day waiting period is built in so that if the citizen doesn't finish their part, the system automatically sends a reminder or closes the application. This prevents the application from being stuck forever without needing a complicated human task system.

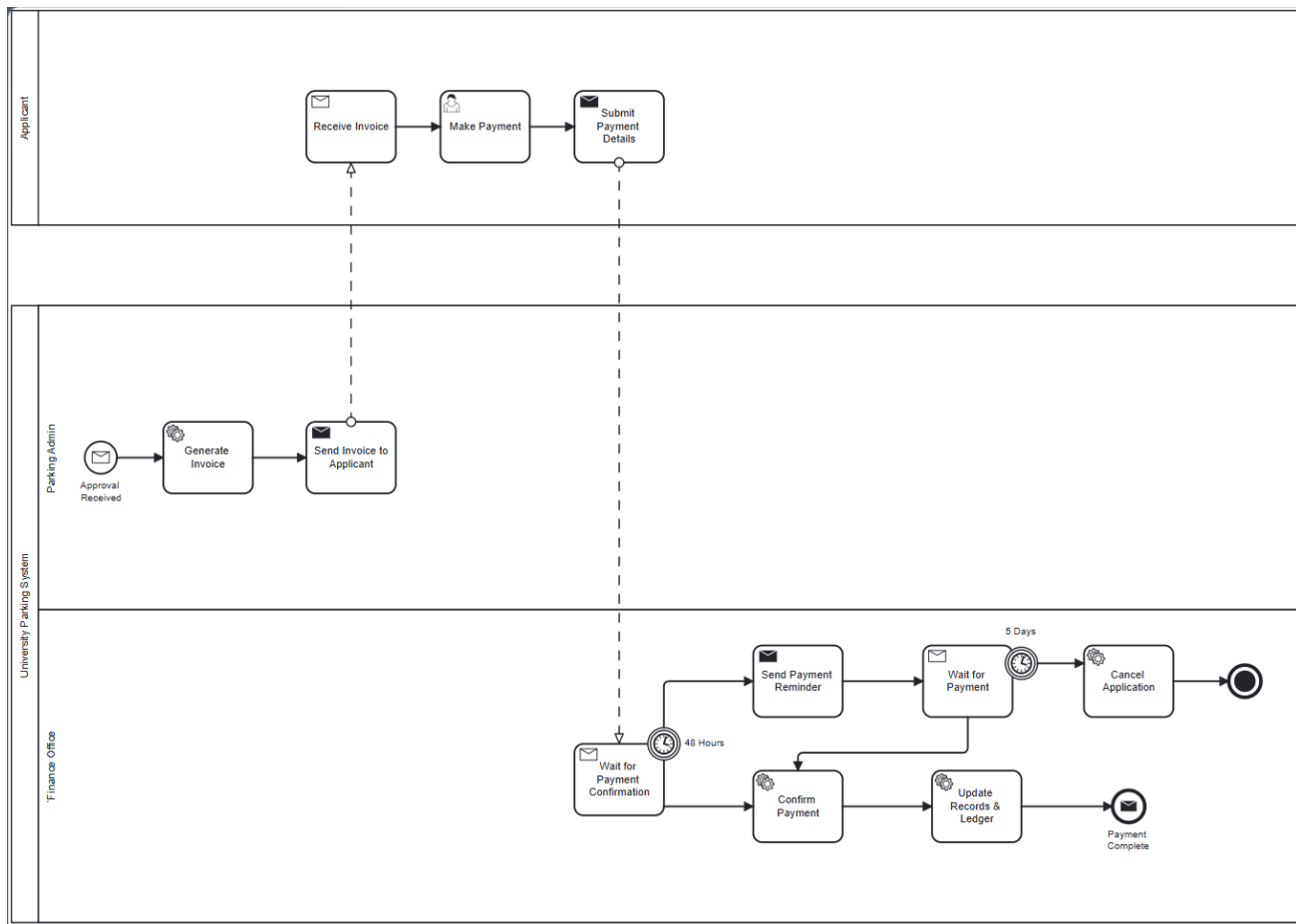


Figure 2: Process 2 - Invoice Generation & Payment Processing

BPMN elements used:

Pools and Lanes: Two pools: Applicant and University Parking System. Inside the University Parking System pool, two lanes: Parking Admin (handles invoice generation) and Finance Office (handles payment waiting and confirmation).

Start Message Event: "Approval received" in the Parking Admin lane, triggered by a message flow from Process 1.

Service Tasks:

- Generate invoice (Parking Admin lane)
- Confirm payment (Finance Office lane)
- Update records & ledger (Finance Office lane)
- Cancel application (Finance Office lane)

Send Tasks:

- Send invoice to applicant (Parking Admin lane).
- Send payment reminder (Finance Office lane).

Receive Tasks:

- Wait for payment (first 48h) (Finance Office lane).
- Wait for payment (final) (Finance Office lane).

Boundary Timer Events:

- On Wait for payment (first 48h): **48 hour timer** and triggers Send payment reminder, then flows to the second Receive Task.
- On Wait for payment (final): **5 day timer** (from original approval) and triggers Cancel application.

Design justification:

The design prioritizes transactional consistency and automated exception handling to reduce administrative overhead. By using two sequential Receive Tasks with interrupting boundary timers (48 hours and 5 days), the system autonomously enforces the client's payment deadlines without using manual monitoring. The 48 hour timer triggers a payment reminder, after which the process moves to a second waiting state then the 5 day timer which is attached to the second Receive Task which cancels the application if payment is still not received ensuring that stale applications do not occupy permit capacity indefinitely. The use of a Terminate End Event on cancellation kills any lingering tokens preventing unwanted side effects such as late payment confirmations. Message flows between the Applicant pool and the Finance Office Lane clearly separate external actions (the applicant paying) from internal system steps (invoice generation, confirmation, ledger update). This architecture creates a 'self cleaning' workflow where the system maintains high data integrity without requiring constant human intervention to reconcile accounts or close abandoned requests.

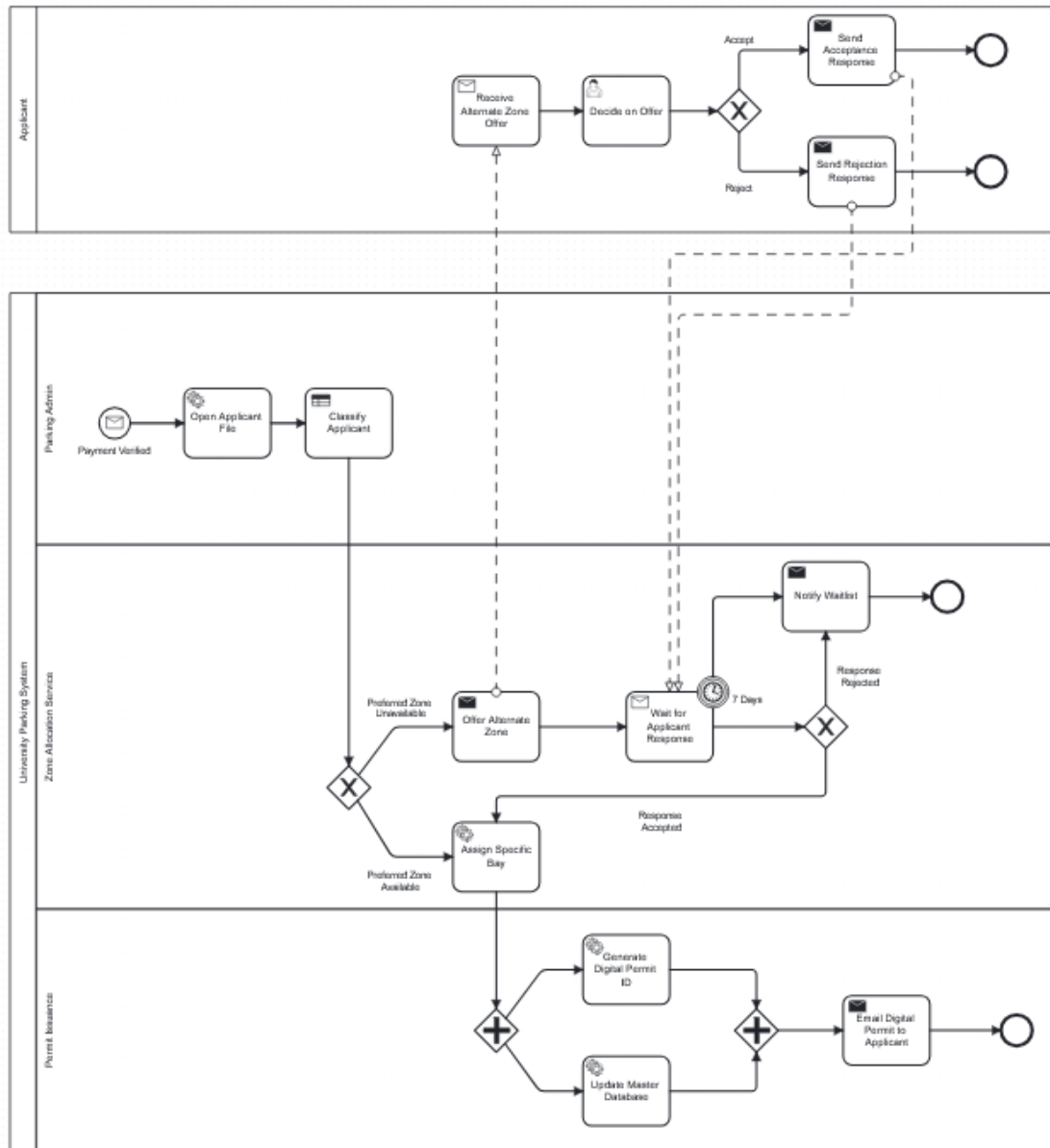


Figure 3 Process 3 - Zone Allocation & Digital Permit Issuance

BPMN elements used:

Pools and Lanes: Two pools: Applicant and University Parking System. Inside the University Parking System pool there are three lanes: Parking Admin, Zone Allocation Service, and Permit Issuance.

Start Message Event: "Payment verified" in the Parking Admin Lane triggered by a message flow from Process 2.

Business Rule Task: Classify applicant determines zone priority (Staff to Zone A, Students to Zones B-D).

Service Tasks:

- Assign specific bay (Zone Allocation Service Lane)
- Generate digital permit ID (Permit Issuance Lane)
- Update master database (Permit Issuance Lane)

Exclusive Gateways:

- "Preferred zone available?" Branches: Available / Unavailable.
- "Response?" Branches: Accepted / Rejected.

Send Tasks:

- Offer alternate zone (to Applicant pool)
- Notify waitlist (on rejection or timeout)
- Email digital permit to applicant

Receive Task: Wait for applicant response (Zone Allocation Service lane). Waits for the applicant's decision on the alternate zone.

Boundary Timer Event: Attached to Wait for applicant response as a **7-day timer**. On timeout it triggers Notify waitlist.

Parallel Gateway: Split and join for concurrent execution of Generate digital permit ID and Update master database.

Design justification:

The design enforces the client's zone allocation priorities (Staff to Zone A, Students to Zones B-D) through a dedicated Business Rule Task making the policy explicit and auditable. When the preferred zone is unavailable, the process does not simply reject the applicant instead, it offers an alternate zone and waits for the applicant's decision via a Receive Task with a 7 day boundary timer. This balances fairness (giving the applicant a choice) with process finality (auto waitlist if no response). The use of a parallel (AND) gateway to concurrently generate the digital permit ID and update the master database reduces processing latency, helping meet the client's target of under one working day. The separation of roles into three lanes (Parking Admin, Zone Allocation Service, Permit Issuance) clearly distinguishes manual oversight (offering alternate zones) from automated system tasks (assigning bays, generating IDs, updating registers). Message flows between the Applicant pool and the system explicitly model the negotiation of an alternate zone, which was only vaguely described in the original proposal.

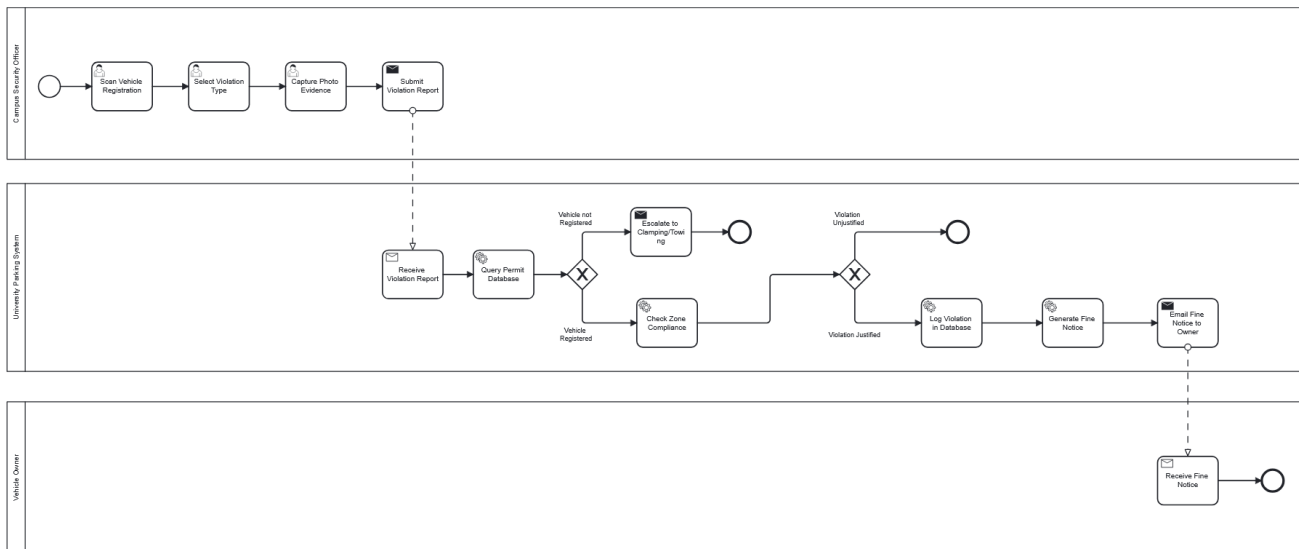


Figure 4: Process 4 - Violation Detection & Fine Issuance

BPMN elements used:

Pools and Lanes: Three pools separate from the external participants (Campus Security Officer, Vehicle Owner) from the University Parking System, clarifying responsibilities between field enforcement, central processing and the vehicle owner.

User Tasks: Manual steps performed by the officer using a mobile device such as Scan vehicle registration, Select violation type, and Capture photo evidence.

Service Tasks: Automated backend operations: Query permit database, Check zone compliance, Log Violation in Database and Generate fine notice.

Send and Receive Tasks: Define handoff points between the officer and the system (Submit violation report and Receive violation report) and the final notification to the owner (Email fine notice and receive a fine notice).

Exclusive Gateways: Handle critical branching. It first checks whether the vehicle is registered (escalate to clamping/towing if not) then checks whether the violation is justified based on permit and zone compliance.

Terminate End Event: Used on the No violation path to end the process when a vehicle is found compliant.

Design justification:

The design focuses on operational agility and automated enforcement consistency. By bridging the gap between field personnel and central databases through Message Flows, the model ensures that enforcement data is transmitted instantly, reducing the 'paperwork lag' common in manual ticketing. The strategic use of Service Tasks for database queries and zone compliance removes subjective human judgment from the verification phase, ensuring that fines are only generated when specific and predefined rules are met. Two exclusive gateways handle critical branching: the first checks

whether the vehicle is registered (escalating unregistered vehicles to clamping/towing), and the second determines if the violation is justified based on permit and zone compliance. If the vehicle is compliant, a 'No Violation' terminate end event ends the process. Most importantly, the escalation path for unregistered vehicles allows the system to prioritize high-risk cases for immediate physical enforcement, while routing standard permit violations through an automated administrative process. This creates a highly scalable system that can handle thousands of daily scans with minimal human oversight.

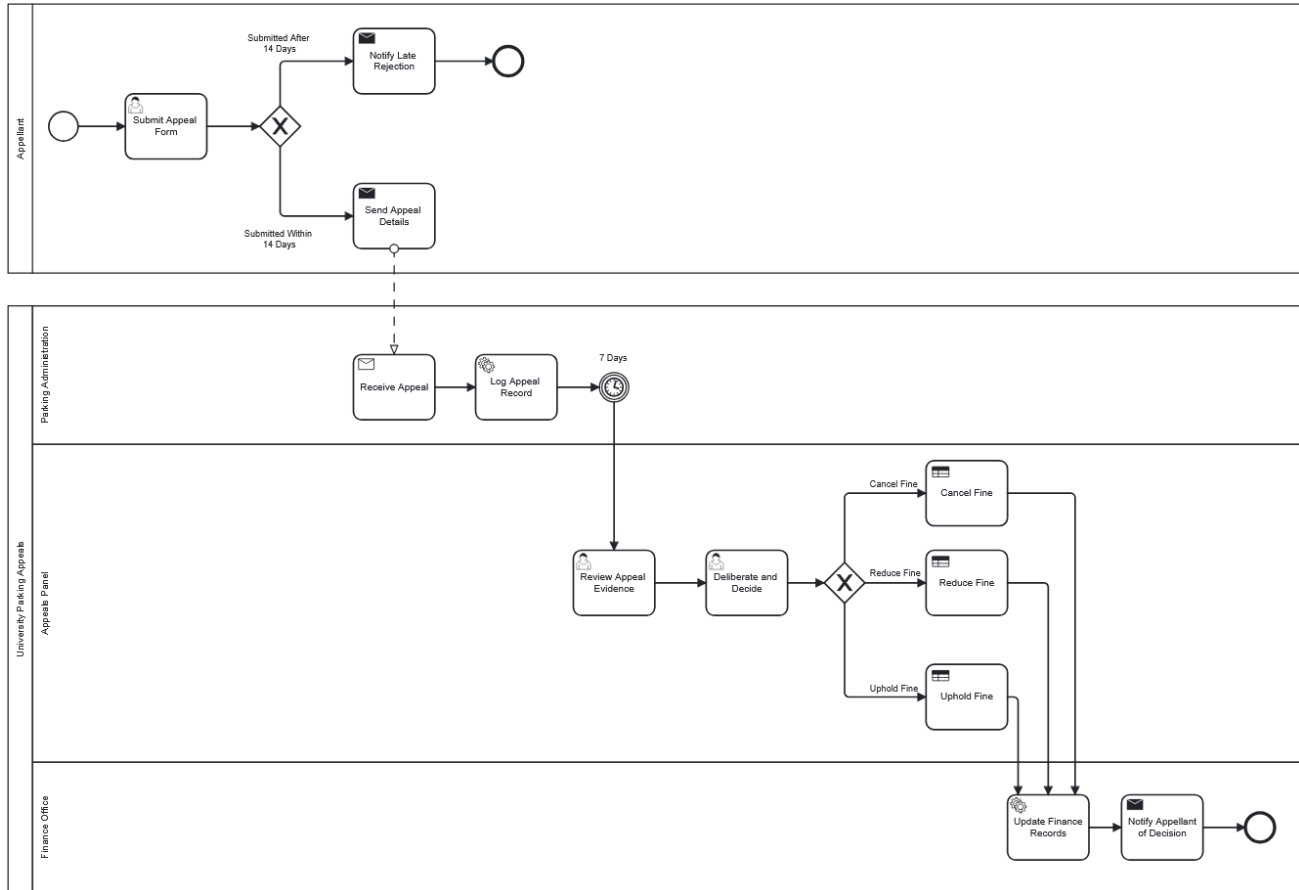


Figure 5: Process 5 - Fine Appeal & Resolution

BPMN elements used:

Pools and Lanes:

Appellant and University Parking Appeals. Inside the Appeals pool, there are three lanes: Parking Administration, Appeals Panel, and Finance Office which separate administrative intake, panel deliberation, and financial record-keeping.

User Tasks: Manual steps requiring human judgment: Submit appeal form (appellant), Review appeal evidence (panel), Deliberate and decide (panel).

Service Tasks: Automated system operations: Log appeal record (Parking Administration), Update finance records (Finance Office).

Business Rule Tasks: Cancel fine, Reduce fine amount, Uphold fine representing decisions that follow standardized university policies rather than ad hoc choices.

Exclusive Gateways: Two critical decision points. First, checking whether the appeal was submitted within the 14-day deadline (Yes/No). Second, the three-way outcome of the panel's deliberation (Cancel / Reduce / Uphold).

Send and Receive Tasks: Handoff of appeal documentation from the Appellant pool to the Parking Administration lane (Send appeal details and Receive appeal), and final decision notification back to the appellant.

Intermediate Catch Timer Event: A clock placed after Log appeal record enforces a mandatory 7-day waiting period (representing the weekly panel meeting schedule) before the Appeals Panel begins the review.

Design justification:

The design emphasises procedural fairness and administrative efficiency by enforcing a strict 14-day submission deadline via a boundary timer on the appellant's submit task. Late appeals are automatically rejected preventing the Appeals Panel from wasting resources on invalid claims. After an appeal is logged, a 7-day intermediate timer event models the client constraint that the panel meets only once per week, explicitly visualising the main source of processing delay. The panel's human-centric review and deliberation are captured in user tasks while the final decision is routed through three business rule tasks (Cancel Fine, Reduce Fine Amount, Uphold Fine) ensuring that outcomes are consistent with university regulations rather than ad-hoc. A service task in the Finance Office lane updates financial records immediately after the decision eliminating manual entry errors. This separation of duties from administrative intake to judicial deliberation to financial execution provides a robust audit trail and protects the integrity of the university's enforcement system.

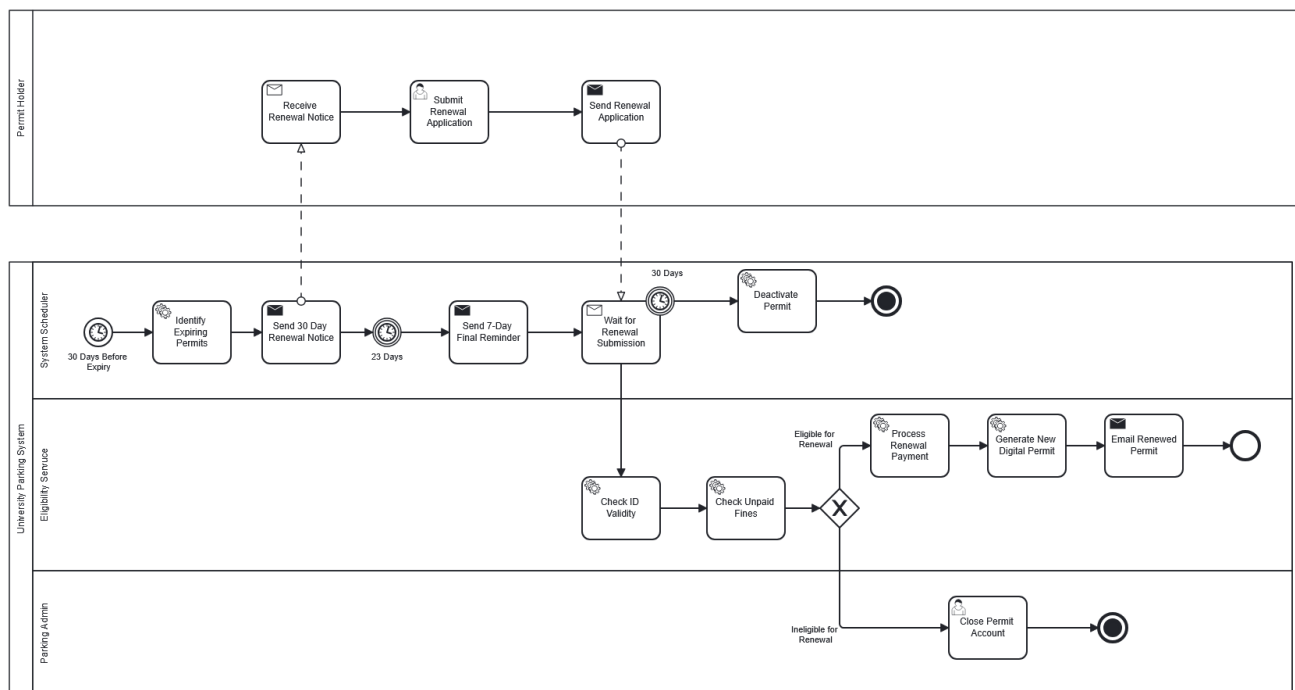


Figure 6: Process 6 - Permit Renewal & Expiry Management

BPMN elements used:

Pools and Lanes: Two pools: Permit Holder and University Parking System. Inside the system pool, there are three lanes: System Scheduler, Eligibility Service and Parking Admin which separate automated timing, eligibility rechecks and administrative closure.

Timer Start Event: 30 days before expiry: Automatically triggers the renewal process for each permit at the appropriate time.

Service Tasks: Automated operations: Identify expiring permits, Check ID validity, Check unpaid fines, Process renewal payment, Generate new digital permit.

Send Tasks: Outgoing notifications: Send 30-day renewal notice, Send 7-day final reminder, Email renewed permit (all with message flows to the Permit Holder pool).

Receive Task: Wait for renewal submission: waits for the permit holder to submit a renewal application.

Boundary Timer Events: Attached to the Receive Task

- 23 days → triggers Send 7-day final reminder, then loops back to the same Receive Task.
- 30 days → triggers Deactivate permit followed by a **Terminate End Event**.

Exclusive Gateway: “Eligible for renewal?”. Branches Yes (proceed to payment and new permit) or No (close account).

User Task: Close permit account: Manual action by Parking Admin when eligibility fails.

Terminate End Events: Used after Deactivate permit and after Close permit account to kill any lingering timers or branches.

Design justification:

The design leverages advanced BPMN elements to balance regulatory compliance with operational efficiency. A timer start event (30 days before expiry) automatically initiates the renewal process eliminating manual scheduling. A Send Task delivers the 30-day renewal notice after which a Receive Task waits for the permit holder's submission. Two interrupting boundary timers on this Receive Task enforce the remaining deadlines: a 23-day timer triggers a 7-day final reminder (sent via a Send Task, then loops back to waiting) while a 30-day timer (expiry) deactivates the permit. This structure ensures that the holder can renew at any time, but the deadlines are never reset incorrectly. When a renewal is submitted, explicit Service Tasks re-check ID validity and unpaid fines followed by an exclusive gateway that directs eligible holders to payment and new permit generation and ineligible cases to account closure. Terminate end events on deactivation and account closure provide hard stops, preventing orphaned process instances or lingering timers. By placing automated checks in the Eligibility Service Lane and manual closure in the Parking Admin Lane, the model maintains clear separation of duties and supports full auditability.

3.2 Discussion and Design Justification

3.2.1 Improvements

- Every deadline in the client's business rules (48h, 5d, 14d, 30d, 7d, expiry) is now modelled as an explicit BPMN timer, making the SLA enforceable by the engine rather than by convention.
- Exception paths omitted in the original proposal such as unresponsive applicant after form return (7-day timer), late appeal submission (14-day boundary timer) and unregistered vehicle escalation are now modelled as explicit boundary events or gateway branches
- Previously implicit external participants (Vehicle Owner) are modelled as collaboration pools with message flows. Payment processing integrates with the university's internal finance system rather than an external gateway.
- The eligibility rule set (ID validity and unpaid fines) is implemented as explicit service tasks in both Process 1 and Process 6 ensuring consistent rule application without hidden complexity.
- Parallel branches (Permit ID generation / master DB update) make concurrency visible and help the client meet the ≤ 1 working day target.

3.2.2 Additional Elements

- Integration with the university's internal finance system for payment processing and corresponding Finance Office integration.
- A dedicated Vehicle Owner pool to close the fine notice message flow.
- An Eligibility Service Lane that performs eligibility checks via explicit Service Tasks (no subprocess).
- Terminate end events on Payment Cancellation and Permit Deactivation to short circuit any lingering timers or reminders.

3.2.3 Assumptions

- The university's SSO (single sign-on) service will authenticate all digital interactions with Applicants and Permit Holders.

- The university's internal finance system provides a real-time payment confirmation API that returns a response within seconds of a successful charge.
- The student/staff database supports real time queries. If unavailable, the system falls back to a manual review task within 24 hours.
- Appeals panel decisions are recorded in the Decision Log synchronously. No asynchronous follow up is required.
- Physical bay availability is accurately mirrored in the digital Zone Register. There is no bay assignment outside the system.
- Zone allocation priorities (Staff are Zone A, Students go to Zones B-D) are fixed policy and can be implemented as a business rule table.

3.2.4 Constraints & Clarifications

- The parking office's Mon-Fri 09:00-17:00 availability constrains any manual review escalations. Where manual intervention is required (e.g., unregistered vehicle escalation), the process uses message flows to hand off asynchronously.
- Budget constraints favored reusing the existing university email system for all notifications rather than building a dedicated messaging service reflected in the use of send tasks rather than custom service tasks.
- Visitor permits, contractor vehicles and disability parking are declared out of scope in the client's proposal; these are not modelled but would extend Process 3 and Process 4 with additional lanes.
- The 500permit physical capacity is treated as a hard constraint enforced by Zone.currentCount, reconciliation jobs are an open clarification we would raise with Group NLSC.

User Interface Design

4. System Design

The data structures required to support the 6 processes are captured in the UML class diagram below. The key classes correspond to each of the data objects exchanged in the BPMN models which are: User, Application, Permit, Zone, Bay, Payment, Invoice, Violation, Fine, Appeals, Appeals Reviewer, and Renewal Notification. Relationships and multiplicity are shown. Composition is used where a parent object exclusively owns its components.

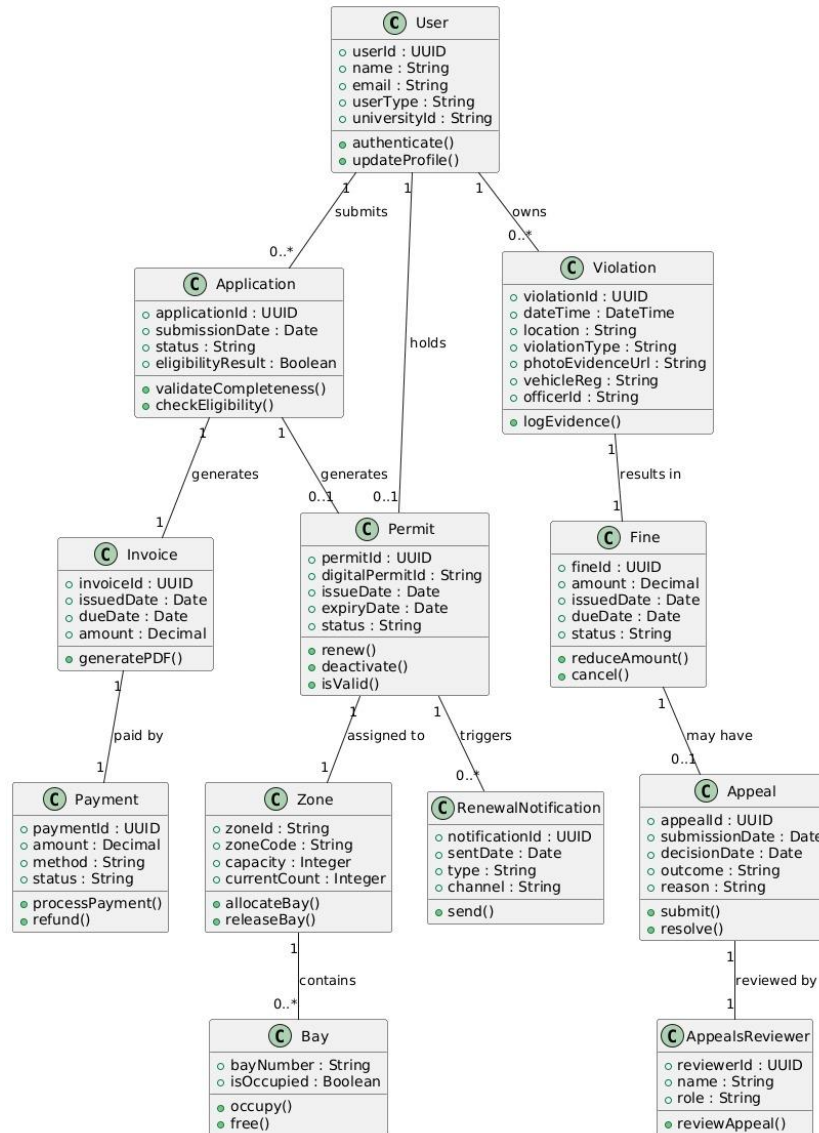


Figure 7: UML Class Diagram supporting Process 1 – Process 6

Consistency with BPMN models:

- Application and Permit persist in the state transitions driven by Process 1 and Process 3.
- Invoice and Payment support the event-based flow in Process 2, including the Pending → Paid / Cancelled state set.
- $\text{Zone.currentCount} / \text{Zone.capacity} = 500$ supports the real time capacity check in Process 3.
- Fine.status enum includes Appealed, consumed by Process 5's gateway.
- Appeal.decision enum mirrors the three-way exclusive gateway in Process 5.
- RenewalNotification captures the 30day and 7day reminder messages sent in Process 6.

5. Optional User Interface Sketches (Bonus)

As a bonus deliverable, our submission includes HTML/CSS wireframes for the three highest traffic user activities: the Applicant's submission form, the Appellant's appeal form, and the Appeals Reviewer's decision form. See UI Wireframes/ in the submission root. These wireframes implement the fields required by DP1's Submit Application Form user task, DP5's Submit Appeal Form user task, and DP5's Appeals Panel deliberation, respectively. They deliberately follow the university's existing web style guide for consistency and to minimize custom development per budget constraint.

Parking Permit Application

Full name:

University ID:

Email:

Role:

Staff ▼

Vehicle plate:

☐ I have no unpaid fines

Submit

Figure 8: Process 1: eligibility & ID checks

Appeal Decision

Fine reference:

Appellant name:

Final decision:

Cancel fine 

If reduced, new amount (\$):

Reasoning:

Record Decision

Process 5: outcome → Cancel/Reduce/Uphold → update finance

Figure 9- Process 5: outcome → Cancel/Reduce/Uphold → update finance

Permit Renewal

Permit holder name:

Current permit ID:

e.g., P-12345

Vehicle plate:

University ID:

☐ I confirm no unpaid fines

Submit Renewal

Figure 10: Process 6 - Permit Renewal & Expiry Management